Vantaku Sai Yashwant

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.43
SRI CHAITANYA JUNIOR KALASALA	Class XII (TSBIE)	2024	97.20%
ST. JOSEPH'S PUBLIC SCHOOL, HYDERABAD	Class X (ICSE)	2022	98.40%

WORK EXPERIENCE

Research Intern - IIT Hyderabad - Professor Shubhadeep Bhattacharjee

- Conducting research on an RRAM-based True Random Number Generator (TRNG).
- Evaluated the randomness of Cadence-based simulation data using the NIST Statistical Test Suite.
- Investigating the impact of device noise on TRNG randomness.
- Exploring its application in Generative Adversarial Networks.

PROJECTS

Pipelined RV32I RISC-V Core - Github

- Designed a **5-stage pipelined RV32I processor** in SystemVerilog
- Integrated hazard detection, full forwarding, stall and flush (without branch prediction).
- Developed a C++/Verilator verification framework to automate execution of the RV32I ISA test suite, reducing manual debugging and enabling regression testing.
- Validated the implementation against the RISC-V test suite, passing **100%** of the tests for the implemented ISA and debugged using GTKwave.

STM32-based Wireless Macropad - Github

- Designed a 2-Layer PCB using the **STM32F103C8T6** microcontroller for a wireless macropad and USB dongle pair in KiCad.
- Learnt about the debugging using **GDB , OPENOCD** and **ST-LINK.**

Modelling of Gate Leakage in MOS using TMM - Github

- Modelled a non linear 1D poisson equilibrium solver to get the potential profile in a MOS capacitor.
- Calculated the transmission coefficient using the transfer matrix method using the potential profile obtained .
- Using the Tsu-Esaki Framework evaluated the equilibrium gate current density. Tsu-Esaki framework , Supply Function modelling

AM Transmitter - Demonstration

- Implemented a 2 stage amplifier electret mic amp with a theoretical gain of around 259
- Using the Colpitts oscillator topology we generated a carrier wave of 1.16MHz
- Level shifted the output of the mic and fed it as the supply of the Colpitts oscillator generating an AM signal

SKILLS

Programming Languages: C, Python, Bash

HDL: System Verilog/Verilog

Simulation: LT-Spice, Cadence Virtuoso

Embedded: OpenOCD/gdb, Arduino IDE, STM32CubeMX

PCB Design: KiCad, Altium

Tools: Git, Make, Linux

Documentation: Latex, Typst

RELEVANT COURSES

Semiconductor Devices:

- **Theory:** Semiconductor Device Fundamentals, Numerical Methods for Device Modelling, Modelling of Nanoscale Transistors, Introduction to CMOS Processing
- **Lab:** Semiconductor Devices Lab

Digital electronics:

- **Theory:** Microprocessor and Computer Architecture, Digital Systems, Productivity Tools for IC design
- **Lab:** Digital Systems Lab, Circuits Lab

Analog and RF electronics:

- **Theory:** Circuits and Network Analysis, Signals and Systems, Probability and Random Processes, Matrix Theory, Analog Electronics, Control Theory, Engineering Electromagnetics
- **Lab:** Circuits Lab, Analog Lab, Electronic System Design Project Lab

POSITIONS OF RESPONSIBILITY

Teaching Assistant - Probability and Random Processes

- Conducted Problem solving sessions
- Helped in evaluation of programming assignments

Teaching Assistant - Engineering Electromagnetics

- Made Problem statements
- Conducted doubt solving sessions